

HypatiaX Paper Trail — Consolidated Audit Summary

Synthesizing: *Supplementary Material A* (routing improvements PDF), *Consolidated Reproducibility & Audit Supplement* (`(supplementary_consolidated_audit)`), and the *Unconfirmed, Withdrawn, and Stale Claims* quarantine document. These three are mutually consistent as a system: the quarantine doc is the ledger, the consolidated audit is the "what's been fixed since" update, and Supp. A is one of the documents being audited.

1. The single most important fact across all three documents

The paper's headline result is reversed, and this is now fully acknowledged in all three documents, not just disclosed once and forgotten.

- Original claim: Hybrid beats Pure LLM, 90.5% vs. baseline (a "+28.3pp" story).
- Root cause found: a "decision-attribution bug" — `(hybrid.success = True)` was set whenever *any* numeric r^2 could be located, without checking whether the sub-method named by `(hybrid.decision)` actually succeeded. This inflated 22/74 (seed 42) near-perfect "successes" that the pipeline's own records show had failed.
- Corrected figure: **45/74 = 60.8%**, *below* Pure LLM's 62.2% on the same run.
- Status across documents: consistent. The consolidated audit's tracker item A confirms this is now the story stated in all 7 places in the main manuscript where 89.2%/90.5% used to conflict (a separate, now-resolved internal inconsistency). The quarantine doc treats the DeFi 1.73×/89.2%/62.2% figures as **"not independently audited by this review"** — i.e., that review scope was Feynman/Core-15, not DeFi — so the DeFi correction's provenance rests on Supp. A / the main paper's own "Hybrid Decision-Attribution Bug" section, not on independent re-derivation by the quarantine reviewers.
- **Caveat not to lose:** the quarantine doc's DeFi section says the *mechanism* (train_r2-based undercounting) originally hypothesized was never confirmed — what's confirmed instead is the *opposite*-direction overcounting bug, and a drafted fix for it **has not been executed against any live run**. So even the corrected 60.8%/62.2% comparison should be read as "root cause identified and demonstrated on 5/5 seeds for one task," not as "recomputed end-to-end and reconfirmed."

Action if citing this paper: cite 60.8% vs 62.2%, not 90.5%. Do not treat the underlying fix as validated by a full corrected re-run — that re-run is still described as pending in the newest document.

2. Runtime / speedup — fully resolved, in the *opposite* direction

All three documents now agree: **no speedup claim survives.**

- Withdrawn: $1.73\times$ (mean) / $1.64\times$ (median) hybrid-faster-than-NN claim, and a 73% runtime-reduction claim.
- What actually happened, per full 5-seed $\times$ 74-task $\times$ 2-split reconstruction (`generate_table1.py`), no hand-edited cells):
 - Aggressive (v3c) split: hybrid **$6.3\times$ – $9.3\times$ slower** than NN, on *every single seed*, no exceptions.
 - PCA-directed split: **$2.5\times$ – $6.1\times$ slower**, and even on seeds routing **zero** tasks to the LLM, hybrid is still $\sim 2.5\times$ slower than NN — meaning there's a fixed hybrid-wrapper overhead independent of LLM calls.
 - The old NN timing (3.0s mean) is $8\text{--}9\times$ larger than what raw files show (0.34–0.42s) — the audit's read is that the old table came from a different, likely older/heavier run, not a computation error.
- Status: **closed/superseded**, tracker item B. Do not resurrect any draft that still asserts a positive speedup or treats this as based on a single seed — the document explicitly warns against exactly that regression.

3. Core-15 ablation ("Table 6") — severe fabrication-pattern, resolved

This is the most damning single item in the whole trail, and all three documents independently converge on it:

Case	Published	Real (<code>merged.json</code>)
Arrhenius (far)	-12.5553 (H), "identical collapse" to PySR	+0.9028 (a success)
Michaelis-Menten (far)	$P=-83,900$, $H=-635$ ("132 $\times$ reduction," abstract headline)	$P=$ +0.647 (success); H is null (never run)
Rate Law / Logistic Growth (H)	$\approx 0.9999\text{--}1.0000$	null , all three ranges
Gravitational Force (H)	finite, moderate-negative	$-\infty$ (genuine crash)
Gravitational Force (P)	finite, moderate-negative	$\approx 0.997\text{--}0.9998$ (near-perfect)
Constant Product (H, med/far)	tracks $P \approx 0.9996$	-5.4×10^{34} / -1.9×10^{46} (collapse)
Portfolio Std Dev (P, far)	-118.4482	1.0000 (success)

Notable: errors run in **both directions** — some published numbers make PySR look artificially worse, others make Hybrid look artificially better — so this isn't simply "thumb on the scale for the headline method," and the paper's domain-selectivity narrative ("Physics/Risk favor HypatiaX, Chemistry/AMM favor PySR-only") was built on these unverified numbers and has been withdrawn along with any Mann-Whitney stats computed from the table.

- Downstream retractions confirmed: abstract's "132× reduction" headline; the Arrhenius/Portfolio "scale-incompatibility" failure-mode narrative; Fig. 9's caption, which independently repeated the same fabricated −83,899 Michaelis-Menten figure as a **third, separately-caught occurrence**; Fig. 18's caption, found to describe a different figure's structure entirely.
 - Status: **closed** — table rebuilt directly from `_merged.json`, no transcription step, cross-checked cell-by-cell in the consolidated audit.
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4. Feynman-30 benchmark — now has a resolved provenance trail, but read the caveats

- Final confirmed figures: **12/30 (40.0%)** random-split, **13/30 (43.3%)** PCA-split. These supersede an earlier 9/30 (withdrawn — traced to a file with `{n_pass: 0, n_total: 0, solve_rate: null}`), duplicated byte-for-byte across 15+ unrelated directories — clearly an unpopulated template, not a real result) and an earlier 18/30 PCA figure (superseded once a 5th defect, missing `asin_of_sin`, was found and patched).
- **Run-date confusion, now disclosed rather than hidden:** at least four distinct PCA-split run dates exist in the repo's history (June 4 — discarded, 0/30 environment failure; June 29 — file never located; June 30 — feeds an unverified, still-untrusted 142/180 pooling artifact; **July 22/23 — the only canonical pair**, confirmed directly by the corresponding author). This matters because a reader who independently opens the June-30 file would see 142/180 and could wrongly conclude the paper is internally inconsistent — it isn't, but only because of this disclosure.
- **The 142/180 figure itself is separately withdrawn:** confirmed to be an exact arithmetic doubling of one 90-record run, not two independent pools.
- **One defect remains permanently unresolvable, not merely unfixed:** placeholder tokens (e.g. `(x0)`, `(x1)`) baked into 4 random-split stored formulas reference variables that were never real inputs — no evaluator patch can recover this; root cause (an upstream serialization step) is not yet located.
- **Gaps to flag if citing per-equation detail:** the 13 already-scored (not rescored) PCA-split equations' individual names/ R^2 values were never captured this round — only the aggregate 10-pass/3-fail split is available. Don't assume you can reconstruct which 10 passed.

- The main paper's old per-equation Feynman table used a **completely different equation set** (Gaussian, Coulomb Force, relativistic momentum, Doppler shift) than the corrected 12/30 and 13/30 runs — it's been relabeled as legacy and two new tables generated from the canonical data replace it.
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5. Nguyen-12 pipeline — model identity resolved, discrepancy narrowed but not closed

- Resolved: the benchmark's actual LLM call path is `hypatia.py`, using `claude-sonnet-5` at temperature **0.25** — not `claude-sonnet-4-6` at 0.3 as the more commonly referenced `LLMConfig` class would suggest. That confusion had a clean explanation: a script sets an `LLM_MODEL` env var that **nothing ever reads** — a "looks wired in but isn't" pattern found twice in this codebase (also true of a `repro.yaml` field).
 - The 91.7%-vs-100% seed-123 discrepancy: **ruled out** as an explanation — seed variance and reduced sample size (cross-seed and within-seed checks both confirm this). **Narrowed** to a ~4-week configuration-migration window (Apr 23–May 22, 2026), plausibly coinciding with a GCP/Kaggle→GitHub Actions infrastructure migration — but this is flagged as worth checking, not confirmed as the cause.
 - Remaining unverified link: `run_all.sh` — the one file that would confirm which script CI actually invokes.
 - Doesn't change any number currently printed in the paper (which already states 33.3%/91.7% in prose), but future re-benchmarking should report a distribution over repeated seed=123 runs, not a single point figure, given confirmed server-side unseeded temperature sampling.
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6. Process/credibility items — do not cite until closed

Item	Status
July 9 CI dashboard ("0 open issues")	Open. Two notebooks directly inspected: one never executed (0/10 cells), one stopped mid-run before its own status cell could declare fixes resolved. <code>build_report.py</code> 's dashboard logic is independent of whether notebooks ran at all — a stale/cached registry could assert "0 open issues" with nothing having run. Code fix (hard-fail on missing/incomplete notebooks) is applied and verified; the actual re-run has not happened. Don't cite this dashboard's conclusions until it has.
Baseline-lock 74/90 vs. 81/90	Resolved, live-confirmed. Threshold bug (0.9999 vs. intended 0.999999) inflated legacy baseline; fresh run shows exactly 74/90, reproduced independently a second time via a separately-discovered Gate C schema bug fix. Cite 74/90.
Gate C <code>_iter_rows</code> schema bug	Confirmed and fixed — a <code>"tests"</code> container key wasn't recognized, silently collapsing every real result file to <code>n_total=0</code> . Verified against synthetic data and a live 12-file real directory, independently reproducing the same 74/90. Porting the fix to two other files still in progress.
Manifest truncation <code>[:N]</code> bug	Fixed in code; not yet fully reconfirmed live — the "5 source files" count is consistent with the fix but hasn't been checked against an actual directory listing.
Second, unreconciled <code>exp2_run.log</code>	Still open — a materially different, non-Feynman-domain benchmark run shares the same filename convention; its relationship to the Feynman results is unresolved.
PCA <code>random_state</code> documentation	Not investigated — tier C, nothing drafted, just flagged.

7. Smaller but real issues, fixed

- Routing-cascade figure vs. architecture figure: **System 1/2/3 numbering conflicts between two figures in the same manuscript** — flagged, genuinely unresolved (not just a caption slip; both figures' own internal legends disagree with each other).
- Seed-sweep caption overgeneralized ("collapses on all seeds," "substantially higher") when the real chart shows a much narrower, less dramatic range — fixed.
- Bibliography: five citations existed only as local `\bibitem`s, not in `references.bib` (added, not fabricated); one citation pointed at the wrong AI-Feynman paper version

(2020 original vs. 2.0 extension) for a claim specifically about the 2.0 figure — both fixed.

- Hardware-variant numbers (Celeron 14/30, Colab 8/30): already correctly caveated in the main text as belonging to the withdrawn 9/30 run; current CI is architecturally incapable of reproducing genuine hardware-variant numbers (uniform `ubuntu-latest` runners) — a capability gap, not a correctness issue, but worth knowing if someone asks for updated hardware numbers.
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8. What this means for citing the paper today

Safe to cite as current:

- Corrected 60.8% (Hybrid) vs. 62.2% (Pure LLM) on the DeFi benchmark.
- Hybrid is $2.5\times$ – $9.3\times$ *slower* than NN, not faster, on every seed/split.
- Rebuilt Core-15 ablation table (from `merged.json`, cross-checked).
- Feynman: 12/30 random-split, 13/30 PCA-split.
- 74/90 legacy baseline-lock figure.
- `claude-sonnet-5` @ temp 0.25 as the actual Nguyen-12 model.

Do not cite:

- 90.5% / "+28.3pp Hybrid advantage" (retracted).
- $1.73\times/1.64\times$ speedup, or the 73% runtime-reduction claim (retracted, reversed).
- Any Table 6 number from the withdrawn draft, or aggregate stats/Mann-Whitney results derived from it.
- The July 9 CI dashboard's "0 open issues" (not yet re-run).
- 142/180 PCA figure, 9/30 Feynman figure, 81/90 baseline (all withdrawn).
- 18/30 PCA-split figure (superseded by 13/30).

Cite with an explicit caveat:

- The DeFi decision-attribution bug's *fix* — root cause confirmed, direction confirmed on 5/5 seeds for one task, but the correction has not been executed against a full live run; treat the 60.8%/62.2% comparison as "diagnosed and directionally validated," not "recomputed end-to-end."
- Nguyen-12 91.7%-vs-100% gap — narrowed to a migration window, not fully explained.
- Planck blackbody's split-divergence (near-miss on random split, catastrophic on PCA split) — the numbers are solid and fully disclosed, but *why* it diverges this sharply remains an open mechanistic question.